

# From DFT to Dielectric Function: A Practical Guide for Engineers

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## 1 Introduction

Dielectric function,  $\epsilon(\omega)$ , characterizes how a material responds to an external electric field and varies with the frequency  $\omega$  of the applied field. This function includes contributions from both electronic excitations and lattice vibrations. In the infrared and terahertz frequency ranges, the dielectric function is significantly influenced by lattice vibrations, particularly optical phonons. Since the dielectric function is intrinsically related to its phonon properties of a material, it is consequently affected by the dynamical stability of its crystal structure [1,2].

Phonons, or quantized vibrational modes, can provide insight into the stability of a crystal structure. When phonon frequencies are calculated, they are derived from the curvature of the potential energy surface (PES) around the equilibrium configuration of the atoms in the material [3,4]. The PES describes the energy of a system as a function of the atomic positions. Phonon frequencies are obtained from the second derivatives of the PES with respect to atomic displacements, which correspond to the curvature of the PES. Positive curvature leads to real phonon frequencies, indicating stable oscillatory motion around the equilibrium position. Negative curvature leads to imaginary phonon frequencies. This implies that small displacements from the equilibrium position will decrease the energy, suggesting that the current configuration is a local maximum or saddle point rather than a minimum. If all phonon frequencies are real (i.e., all curvatures are positive), the structure is dynamically stable.

Density Functional Perturbation Theory (DFPT) and the finite displacement method are two primary approaches for calculating phonon properties in crystals [5]. Phonon calculation in the finite displacement method involves calculating forces in Density Functional Theory (DFT) and determining the second derivatives of the energy using finite differences. This method typically requires constructing supercells to capture long-wavelength phonons, which can be computationally expensive. DFPT does not require the construction of supercells. It calculates the response to a finite wave vector within the primitive cell, which reduces the computational cost compared to the finite displacement method [6]. In DFPT, the idea is to apply a small perturbation to the system and then calculate the first-order response of the electron density and potential to this perturbation. This approach allows us to compute the second derivatives of the total energy, which are related to the force constants and phonon properties. The second derivative of the energy with respect to atomic positions in DFPT can be derived by considering the total energy expression in DFT and perturbing the system with respect to atomic displacements.

The starting point of any phonon calculation is the Born-Oppenheimer approximation, which allows us to separate the electron and nuclear degrees of freedom. This simplifies the problem of many-body quantum systems by separating the electronic and nuclear motions. It assumes that the nuclei are much heavier and move much slower than the electrons, so the electrons adjust instantaneously to the positions of the nuclei. The Hamiltonian  $H$  of a system, when applying the Born-Oppenheimer approximation, can be expressed as [7]:

$$H = - \sum_I \frac{\hbar^2}{2M_I} \frac{\partial^2}{\partial \mathbf{R}_I^2} + E(\{\mathbf{R}\}) \quad (1)$$

Where,  $\hbar$  reduced Planck constant,  $M_I$  mass of the  $I$ -th nucleus. The second derivative with respect to the displacement of the  $I$ -th nucleus represents the kinetic energy of the nuclei.  $E(\{\mathbf{R}\})$  the potential energy of the system as a function of the nuclear positions  $\mathbf{R}$ . This equation 1 represents the electronic energy obtained from solving a electronic Schrödinger equation for fixed nuclear positions. The total energy  $E$  can be expanded in a Taylor series with respect to:

$$E(\mathbf{R}) = E_0 + \sum_I \left( \frac{\partial E}{\partial \mathbf{R}_I} \right)_{\mathbf{R}=0} \mathbf{R}_I + \frac{1}{2} \sum_I \left( \frac{\partial^2 E}{\partial \mathbf{R}_I \partial \mathbf{R}_J} \right)_{\mathbf{R}=0} \mathbf{R}_I \mathbf{R}_J + \dots \quad (2)$$

$E_0$  is the total energy at the equilibrium position. The equilibrium geometry of the system is given by the condition that the forces acting on individual nuclei vanish:

$$F_I = - \frac{\partial E(\mathbf{R})}{\partial \mathbf{R}_I} = 0 \quad (3)$$

the vibrational frequencies  $\omega$ , are determined by the eigenvalues of the Hessian of the ground state energy of a system of electrons moving in the field of fixed nuclei with coordinates  $\mathbf{R}$ , scaled by the nuclear masses [5]:

$$\det \left| \frac{1}{\sqrt{M_I M_J}} \frac{\partial^2 E(\mathbf{R})}{\partial \mathbf{R}_I \partial \mathbf{R}_J} - \omega^2 \right| = 0 \quad (4)$$

where  $M_I$  and  $M_J$  are the masses of atoms  $I$  and  $J$ , respectively. The Hessian matrix is usually called the matrix of the inter-atomic force constants(IFC). The forces  $\mathbf{F}_I$  can be calculated by applying the Hellmann–Feynman (HF) theorem to the Born–Oppenheimer Hamiltonian  $H_{BO}$

$$\mathbf{F}_I \equiv - \frac{\partial E(\{\mathbf{R}\})}{\partial \mathbf{R}_I} = - \left\langle \Psi_{\{R\}} \left| \frac{\partial H_{BO}}{\partial \mathbf{R}_I} \right| \Psi_{\{R\}} \right\rangle \quad (5)$$

$\Psi_{\{R\}}$  is the eigenfunction of  $H_{BO}$  corresponding to the  $E(\{\mathbf{R}\})$  eigenvalue of Schrödinger equation.

$$\hat{H}_{BO} \Psi_{\{\mathbf{R}\}} = E(\mathbf{R}) \Psi_{\{\mathbf{R}\}} \quad (6)$$

The BO Hamiltonian depends on  $\mathbf{R}$  via the electron-ion interaction that couples to the electronic degrees of freedom only through the electron charge density. The HF theorem states in this case that [7]:

$$\mathbf{F}_I = - \int n(\mathbf{r}) \frac{\partial V_{ext}(\mathbf{r})}{\partial \mathbf{R}_I} d\mathbf{r} - \frac{\partial E_N(\{\mathbf{R}\})}{\partial \mathbf{R}_I} \quad (7)$$

$V_{ext}(\mathbf{r})$  is the attractive interaction between electrons and nuclei:

$$V_{ext}(\mathbf{r}) = - \sum_{iI} \frac{Z_I e^2}{|\mathbf{r}_i - \mathbf{R}_I|} \quad (8)$$

where,  $Z_I$ ,  $\mathbf{R}_I$  being the charge and coordinate of the  $I$ th nucleus respectively and  $\mathbf{r}_i$  is the coordinate of the  $i$ th electron.  $E_N(\{\mathbf{R}\})$  is the nuclear electrostatic energy:

$$E_N(\{\mathbf{R}\}) = \frac{e^2}{2} \sum_{I \neq J} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|} \quad (9)$$

$e$  is the electron charge. IFCs can be calculated as finite differences of Hellmann–Feynman forces for small finite displacements of atoms around the equilibrium positions (frozen phonon technique). An alternative technique is the direct calculation of IFCs using density-functional perturbation theory (DFPT). An explicit expression for the IFCs can be obtained by differentiating the forces with respect to nuclear coordinates [7],

$$\Phi_{IJ}(\mathbf{R}) = \frac{\partial^2 E(\{\mathbf{R}\})}{\partial \mathbf{R}_I \partial \mathbf{R}_J} = \int \frac{\partial n(\mathbf{r})}{\partial \mathbf{R}_I} \frac{\partial V_{ext}(\mathbf{r})}{\partial \mathbf{R}_J} d\mathbf{r} + \delta_{IJ} \int n(\mathbf{r}) \frac{\partial^2 V_{ext}(\mathbf{r})}{\partial \mathbf{R}_I \partial \mathbf{R}_J} d\mathbf{r} + \frac{\partial^2 E_N(\{\mathbf{R}\})}{\partial \mathbf{R}_I \partial \mathbf{R}_J} \quad (10)$$

where,  $\Phi_{IJ}(\mathbf{R})$  represents the force constant matrix elements between atoms  $I$  and  $J$  at positions  $\mathbf{R}$ .  $\delta_{IJ}$  is the Kronecker delta, which equals 1 if  $I = J$  and 0 otherwise. The calculation of the IFCs thus requires the calculation of the ground-state electron charge density,  $n$ , as well as of its linear response to a distortion of the nuclear geometry,  $\partial n(\mathbf{r})/\partial \mathbf{R}_I$ . Within the framework of density-functional theory (DFT), the ground state charge density  $n_{\mathbf{R}}$  can be evaluated. Followings are the details of how we can find the parameters in equation 10 using DFT and DFPT.

## 2 Density function theory

The starting point for DFPT is the ground-state electronic structure of the system, which includes the ground-state charge density  $n(\mathbf{r})$  and the ground-state potential  $V_{ext}(\mathbf{r})$ . These quantities are obtained using DFT. The final expression of the total energy  $E(\{\mathbf{R}\})$  in the framework of the DFT consists of five energy terms [8].

$$E(\{\mathbf{R}\}) = E_{kin}^{non}[\Phi_n(\mathbf{r})] + E_{ext}[n(\mathbf{r})] + E_H[n(\mathbf{r})] + E_{xc}[n(\mathbf{r})] + E_N(\{\mathbf{R}\}) \quad (11)$$

All terms except kinetic energy and nuclear electrostatic energy are written exclusively in terms of electron density, and kinetic energy is written as a functional of the noninteracting orbitals. Here, the repulsive interaction energy between the nuclei  $E_N(\{\mathbf{R}\})$  is added as a constant within the Born–Oppenheimer approximation. In a noninteracting system of the Kohn–Sham scheme, however, kinetic energy can be expressed in summation form with the KS orbitals  $\Phi_n$ ,

$$E_{kin}^{non}[n(\mathbf{r})] = -2 \frac{\hbar^2}{2m} \sum_{n=1}^{N/2} \int \Phi_n^*(\mathbf{r}) \frac{\partial^2 \Phi_n}{\partial \mathbf{r}^2} d\mathbf{r} \quad (12)$$

Although orbitals are involved here, it is directly related to electron density via Equation 13,

$$n(\mathbf{r}) = 2 \sum_{n=1}^{N/2} |\Phi_n(\mathbf{r})|^2 \quad (13)$$

The external energy  $E_{ext}$ , expectation value of  $V_{ext}$ , is determined by quantum manipulation. The external potential,  $V_{ext}$ , comes from the interaction between an electron and other nuclei [8].

$$E_{ext}[n(\mathbf{r})] = \int \Phi_n^*(\mathbf{r}) V_{ext} \Phi_n(\mathbf{r}) d\mathbf{r} = \int V_{ext} n(\mathbf{r}) d\mathbf{r} \quad (14)$$

The Hartree energy  $E_H$ , Coulomb energy for the interaction between an electron at  $\mathbf{r}$  and the mean electron density at  $\mathbf{r}'$  in a mean-field approximation, can be expressed as:

$$E_H[n(\mathbf{r})] = \frac{e^2}{2} \int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}' \quad (15)$$

The final term is XC energy  $E_{xc}$ , which consists of all quantum effects, and is approximated in terms of electron density  $n(\mathbf{r})$ . Having an accurate XC energy becomes a key factor for the successful operation of the DFT. The local-density approximation (LDA), is surprisingly good for silicon system and we considered this approximation in our DFT model.

The Kohn-Sham (KS) orbitals  $\Phi_n(\mathbf{r})$ , are the solutions of the KS equation [8]:

$$\hat{H}_{KS}\Phi_n(\mathbf{r}) \equiv \left[ -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial \mathbf{r}^2} + V_{eff}(\mathbf{r}) \right] \Phi_n(\mathbf{r}) = \varepsilon_n \Phi_n(\mathbf{r}) \quad (16)$$

where,  $\varepsilon_n$  is the eigenvalue (energy) of the  $n$ -th Kohn-Sham (KS) orbital.  $\hat{H}_{KS}$  is the Hamiltonian for an electron under an effective potential  $V_{eff}$ :

$$V_{eff}(\mathbf{r}) = V_{ext}(\mathbf{r}) + e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + V_{xc}(\mathbf{r}) \quad (17)$$

$V_{xc}$  – the exchange-correlation potential – is the functional derivative of the exchange-correlation energy. It is determined by [5, 7]:

$$V_{xc}(\mathbf{r}) \equiv \frac{\delta E_{xc}}{\delta n(\mathbf{r})} \quad (18)$$

In a pseudopotential scheme each nucleus is thought to be lumped together with its own core electrons in a frozen ion which interacts with the valence electrons through a smooth pseudopotential,  $V_{ext}(\mathbf{r})$ .

The Kohn-Sham equation 16 is a nonlinear differential equation and hence we need to solve it self-consistently. The nonlinearity in the Kohn-Sham (KS) equations arises primarily from the dependence of the effective potential  $V_{ext}$  on the electron density  $n(\mathbf{r})$  specifically, Hartree potential  $V_H$  and exchange-correlation potential  $V_{xc}$ . The workflow of the self-consistent field (SCF) can be summarized as the following:

1. Start with an initial guess for the electron density  $n^{(0)}(\mathbf{r})$ .
2. Compute the effective potential  $V_{eff}[n^{(0)}](\mathbf{r})$  using the initial guess.
3. Solve the KS equations using  $V_{eff}[n^{(0)}](\mathbf{r})$  to get the Kohn-Sham orbitals  $\Phi_n^{(1)}(\mathbf{r})$  and eigenvalues  $\varepsilon_n^{(1)}$ .
4. Construct a new electron density  $n^{(1)}(\mathbf{r})$  from the obtained orbitals using equation 13.
5. Compare the new electron density  $n^{(1)}(\mathbf{r})$  with the old density  $n^{(0)}(\mathbf{r})$ . If the difference is below a predefined tolerance, the process is complete. Otherwise, update the density:

$$n^{(n+1)}(\mathbf{r}) = n^{(n)}(\mathbf{r}) + \lambda \left( n^{(n)}(\mathbf{r}) - n^{(n-1)}(\mathbf{r}) \right)$$

where  $\lambda$  is a mixing parameter to aid convergence.

6. Repeat steps 2 to 5 until the electron density converges, meaning  $n^{(n)}(\mathbf{r}) \approx n^{(n-1)}(\mathbf{r})$  within the specified tolerance.

This method ensures that the electron density and potential are in mutual equilibrium, leading to accurate calculations of various physical properties. In periodic systems the index  $n$  running over occupied states can be thought as a double label,  $n \equiv \{v, k\}$ , where  $v$  indicates the set of valence bands, and  $k$  is a wave-vector belonging to the first Brillouin zone (BZ).

### 3 Density-Functional Perturbation Theory

In the context of DFPT, the focus is on how the ground-state effective potential  $V_{eff}(\mathbf{r})$  and the electronic structure respond to small perturbations, such as atomic displacements. The charge-density linear response can be evaluated by linearizing Eqs. 13, 16 and 17 with respect to derivatives of KS orbitals, density, and potential, respectively. Linearization of Eq. 13 leads to:

$$\frac{\partial n(\mathbf{r})}{\partial \mathbf{R}_I} = 4 \text{Re} \sum_{n=1}^{N/2} \int \Phi_n^*(\mathbf{r}) \frac{\partial \Phi_n(\mathbf{r})}{\partial \mathbf{R}_I} \quad (19)$$

This equation expresses the response of the charge density to nuclear displacements in terms of the derivatives of Kohn-Sham (KS) orbitals  $\Phi_n(\mathbf{r})$ . The derivatives of the KS orbitals,  $\partial \Phi_n(\mathbf{r})/\partial \mathbf{R}_I$ , are obtained from linearization of Eqs. 16 and 17:

$$\left( \hat{H}_{KS} - \varepsilon_n \right) \frac{\partial \Phi_n(\mathbf{r})}{\partial \mathbf{R}_I} = - \left( \frac{\partial V_{eff}(\mathbf{r})}{\partial \mathbf{R}_I} - \frac{\partial \varepsilon_n}{\partial \mathbf{R}_I} \right) \Phi_n(\mathbf{r}) \quad (20)$$

where

$$\frac{\partial V_{eff}(\mathbf{r})}{\partial \mathbf{R}_I} = \frac{\partial V_{ext}(\mathbf{r})}{\partial \mathbf{R}_I} + e^2 \int \frac{1}{|\mathbf{r} - \mathbf{r}'|} \frac{\partial n(\mathbf{r}')}{\partial \mathbf{R}_I} d\mathbf{r}' + \int \frac{\delta V_{xc}}{\delta n(\mathbf{r}')} \frac{\partial n(\mathbf{r}')}{\partial \mathbf{R}_I} d\mathbf{r}' \quad (21)$$

is the first-order derivative of the self-consistent potential, and

$$\frac{\partial \varepsilon_n}{\partial \mathbf{R}_I} = - \left\langle \Phi_n \left| \frac{\partial V_{eff}}{\partial \mathbf{R}_I} \right| \Phi_n \right\rangle \quad (22)$$

is the first-order derivative of the KS eigenvalue,  $\varepsilon_n$ .

Equation 20 describes how the KS orbitals change in response to a perturbation in the nuclear positions, and it is solved iteratively as part of the self-consistent linear response procedure in DFPT. For each perturbed configuration, solve the linearized Kohn-Sham equation 20 to find the first-order changes in the KS orbitals  $\partial \Phi_n(\mathbf{r})/\partial \mathbf{R}_I$ . Calculate the change in charge density  $\partial n(\mathbf{r})/\partial \mathbf{R}_I$  using the obtained orbital derivatives. Compute the first-order derivative of the self-consistent potential  $\partial V_{eff}(\mathbf{r})/\partial \mathbf{R}_I$ . Repeat the process of solving the linearized equations, updating the charge density and potential until self-consistency is achieved. This means the changes in the charge density and potential response converge. Thus, the ground-state DFT calculation provides the necessary inputs, and the DFPT calculation determines the system's response to perturbations.

Now, we can use equation 10 to find the IFCs as we have the necessary quantities from the ground-state DFT and DFPT calculations. The IFCs provide crucial information about the forces between atoms in response to small displacements, which is essential for understanding phonon properties and lattice dynamics. In a crystal, atoms are

arranged in a periodic lattice, phonons exhibit wave-like properties and are characterized by wavevectors  $\mathbf{q}$ . These wavevectors lie within the first Brillouin zone of the reciprocal lattice. To analyze phonons in a periodic crystal, the system is often treated as an infinite periodic array, allowing for the description of phonons as traveling waves with specific wavevectors that capture the vibrational modes of the entire crystal. In a periodic system, the IFCs  $\Phi_{IJ}(\mathbf{R})$  in real space can be transformed to the reciprocal space dynamical matrix  $D_{IJ}(\mathbf{q})$  using a Fourier transform [9].

$$D_{IJ}(\mathbf{q}) = \frac{1}{\sqrt{M_I M_J}} \sum_{\mathbf{R}} \Phi_{IJ}(\mathbf{R}) e^{i\mathbf{q} \cdot \mathbf{R}} \quad (23)$$

The IFCs describe the force on atom  $I$  in a unit cell due to a displacement of atom  $J$ .  $\mathbf{R}$  runs over all lattice vectors. To obtain the phonon dispersion relations, the dynamical matrix  $D_{IJ}(\mathbf{q})$  is diagonalized at each  $\mathbf{q}$ -point. Accurate calculation of phonon properties requires sampling the Brillouin zone at a sufficient number of  $\mathbf{q}$ -points. To verify the IFCs, we plot the phonon dispersion (Figure 1) and compare it with literature data [10]. Phonon dispersion curves  $\omega(\mathbf{q})$  are plotted along high-symmetry directions in the Brillouin zone, providing crucial information about the vibrational properties of the crystal. Upon validation of the phonon properties, we proceed to calculate the dielectric function, which is described in the next section.

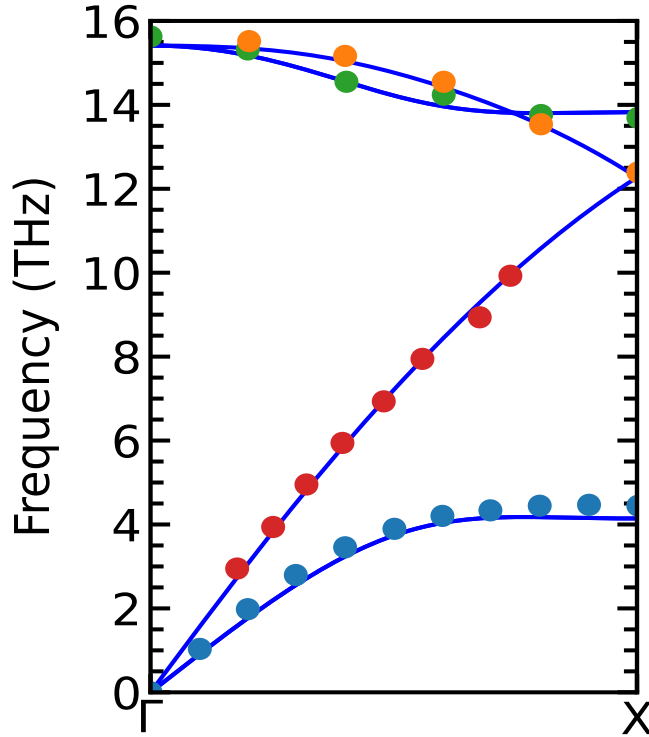


Figure 1: Phonon dispersion at 0 K comparison with S. Wei et al. [10] Line plot is generated by the present study.

To incorporate temperature effects into DFPT calculations, we account for the temperature-dependent lattice constants and atomic positions. Specifically, we utilize a temperature-dependent atomic positions derived from theoretical models, Special Displacement Method

[11,12] , which capture the thermal effect on atomic positions. Additionally, we employ Car-Parrinello Molecular Dynamics (CPMD) simulations [13] to determine the lattice constants at various temperatures, allowing us to precisely model the atomic configurations under thermal conditions. By integrating these temperature-dependent parameters into our DFPT framework, we accurately assess how thermal vibrations influence the phonon properties and electronic structure, leading to a comprehensive understanding of the material's behavior at finite temperatures.

## 4 Dielectric tensor

Density Functional Theory (DFT) calculations (section 2) provide the essential electronic structure information needed to determine the dielectric function of solid materials. By solving the Kohn-Sham equation 16, DFT yields the eigenvalues  $\varepsilon_{\mathbf{k},n}$  and eigenvectors  $\Phi_{\mathbf{k},n}(\mathbf{r})$  corresponding to the electronic states in the crystal. Here,  $\mathbf{k}$  represents the wave vector in the Brillouin zone, and  $n$  denotes the band index. These eigenvalues represent the energy levels of electrons in the solid, while the eigenvectors describe their probability distributions. Utilizing this information, one can compute the dielectric tensor  $\epsilon_{\alpha,\beta}(\omega)$ , which characterizes the material's response to external electric fields and is crucial for understanding various optical properties, including absorption, reflectivity, and energy loss spectra. It can be separated into real and imaginary parts, each representing different physical phenomena. This part is related to the material's ability to polarize in response to the electric field. Real part of the dielectric tensor reflects the material's dielectric constant, which affects how the material's electric field is modified by the presence of the material. However, imaginary part is related to the absorption of electromagnetic waves by the material. It represents how the material loses energy and related to the material's absorption coefficient and the attenuation of the wave as it propagates through the material.

**Imaginary Part of the Dielectric Tensor** The imaginary part of the dielectric tensor,  $\epsilon_{\alpha,\beta}^{imag}(\omega)$ , can be derived from perturbation theory with adiabatic switching on [14]:

$$\epsilon_{\alpha,\beta}(\omega) = 1 + \frac{4\pi e^2}{\Omega N_{\mathbf{k}} m^2} \sum_{n,n'} \sum_{\mathbf{k}} \frac{\hat{\mathbf{M}}_{\alpha,\beta}}{(\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n})^2} \left\{ \frac{f(\varepsilon_{\mathbf{k},n})}{\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n} + \hbar\omega + i\hbar\Gamma} + \frac{f(\varepsilon_{\mathbf{k},n})}{\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n} - \hbar\omega - i\hbar\Gamma} \right\} \quad (24)$$

Here:

- $\omega$  is the frequency of the perturbation (e.g., an external electromagnetic field).
- $e$  is the electron charge.
- $\Omega$  is the volume of the crystal.
- $N_{\mathbf{k}}$  is the number of  $\mathbf{k}$ -points in the Brillouin zone sampling.
- $m$  is the electron mass.
- $n$  and  $n'$  are band indices.
- $\mathbf{k}$  is the wavevector in the Brillouin zone.
- $\varepsilon_{\mathbf{k},n'}$  and  $\varepsilon_{\mathbf{k},n}$  are the energies of the electronic states with wavevector  $\mathbf{k}$  and band indices  $n$  and  $n'$ , respectively.

- $f(\varepsilon_{\mathbf{k},n})$  is the Fermi-Dirac distribution function, indicating the occupation of the electronic state with energy  $\varepsilon_{\mathbf{k},n}$ .
- $\hat{\mathbf{M}}_{\alpha,\beta}$  is a matrix element related to the momentum operator.
- $\hbar$  is the reduced Planck constant.
- $\Gamma$  is a small adiabatic parameter, representing broadening and ensuring energy conservation in a real physical system.

Simplifying the above equation in the limit where  $\Gamma$  approaches zero, we get:

$$\epsilon_{\alpha,\beta}^{imag}(\omega) = \frac{4\pi e^2}{\Omega N_{\mathbf{k}} m^2} \sum_{n,n'} \sum_{\mathbf{k}} \frac{\hat{\mathbf{M}}_{\alpha,\beta} f(\varepsilon_{\mathbf{k},n})}{(\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n})^2} [\delta(\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n} + \hbar\omega) + \delta(\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n} - \hbar\omega)] \quad (25)$$

This expression highlights that the imaginary part  $\epsilon_{\alpha,\beta}^{imag}$  is directly related to electronic transitions within the material. The delta functions enforce energy conservation for photon absorption and emission processes. It signifies that only specific transitions contribute to the imaginary part of the dielectric function, where the energy difference between states matches the photon energy  $\hbar\omega$ .

The imaginary part of the dielectric tensor represents the material's absorption characteristics. When an electromagnetic wave with frequency  $\omega$  interacts with the material, it can cause electronic transitions between states  $|\mathbf{k}, n\rangle$  and  $|\mathbf{k}, n'\rangle$ . The term  $\hat{\mathbf{M}}_{\alpha,\beta}$  represents the transition matrix element that determines the probability of these transitions occurring.

The Dirac delta function represents an idealized, infinitely sharp condition for energy matching. In practice, due to broadening effects (e.g., thermal effects, impurities, and finite lifetimes), transitions can occur over a range of photon energies near  $\hbar\omega$ . The transition from the idealized case with Dirac delta functions to a more realistic Drude-Lorentz form takes into account the intrinsic broadening of excited states due to interactions with the electromagnetic field and other factors. In the Drude-Lorentz model,  $\Gamma$  is a small but non-zero parameter that represents this broadening. It modifies the response function from the ideal Dirac delta case to a more realistic Lorentzian profile, where transitions occur over a range of photon energies near the resonance condition [14].

$$\epsilon_{\alpha,\beta}^{imag}(\omega) = \frac{4\pi e^2}{\Omega N_{\mathbf{k}} m^2} \sum_{n,\mathbf{k}} \frac{df(\varepsilon_{\mathbf{k},n})}{d\varepsilon_{\mathbf{k},n}} \frac{\eta\omega \hat{\mathbf{M}}_{\alpha,\beta}}{\omega^4 + \eta^2\omega^2} + \frac{8\pi e^2}{\Omega N_{\mathbf{k}} m^2} \sum_{n \neq n'} \sum_{\mathbf{k}} \frac{\hat{\mathbf{M}}_{\alpha,\beta}}{\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n}} \left[ \frac{\Gamma\omega}{[(\varepsilon_{\mathbf{k},n'} - \varepsilon_{\mathbf{k},n})^2 - \omega^2]^2 + \Gamma^2\omega^2} \right] f(\varepsilon_{\mathbf{k},n}) \quad (26)$$

Here,  $\eta$  represents the plasma frequency, and this term describes the frequency-dependent response of free carriers. The Dirac delta function implies perfect energy conservation and infinite lifetimes of states, which is an idealization. However, Drude-Lorentz model incorporates intrinsic broadening and finite lifetimes of excited states, providing a more realistic description of the material's response to electromagnetic fields.

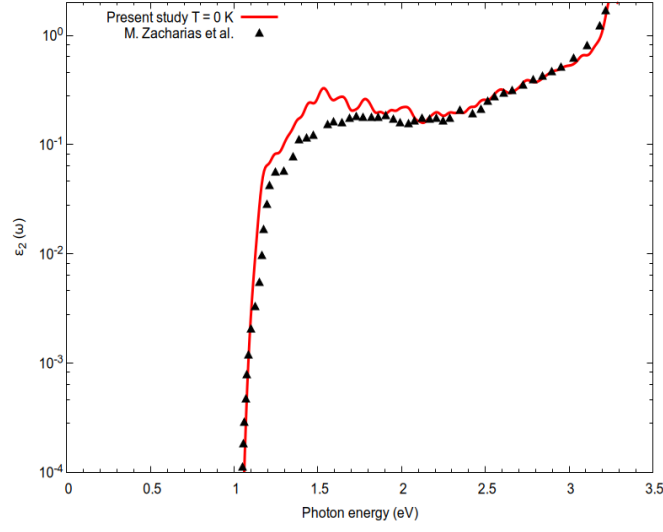


Figure 2: Imaginary part of dielectric function at 0 K comparison with M. Zacharias et al. [12]

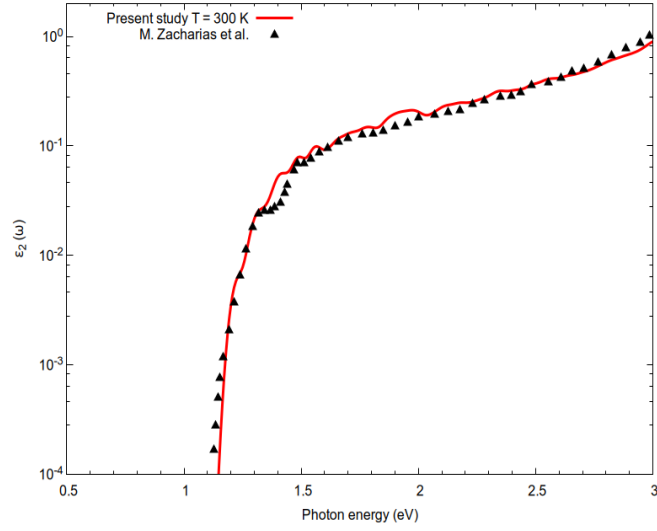


Figure 3: Imaginary part of dielectric function at 300 K comparison with M. Zacharias et al. [12]

**Real Part of the Dielectric Tensor** The real part of the dielectric tensor,  $\epsilon_{\alpha,\beta}^{real}(\omega)$ , is obtained through the Kramers-Kronig relation [14]:

$$\epsilon_{\alpha,\beta}^{real}(\omega) = 1 + \frac{2}{\pi} \int_0^{\infty} \frac{\omega' \epsilon_{\alpha,\beta}^{imag}(\omega')}{\omega'^2 - \omega^2} d\omega' \quad (27)$$

$$\epsilon_{\alpha,\beta}^{real}(\omega) = 1 + \frac{2}{\pi} \int_0^{\infty} \frac{\omega' \epsilon_{\alpha,\beta}^{imag}(\omega')}{\omega'^2 - \omega^2} d\omega' \quad (28)$$

$$\epsilon_{\alpha,\beta}^{real}(\omega) = 1 - \frac{4\pi e^2}{\Omega N_{\mathbf{k}} m^2} \sum_{n,\mathbf{k}} \frac{df(\epsilon_{\mathbf{k},n})}{d\epsilon_{\mathbf{k},n}} \frac{\omega^2 \hat{\mathbf{M}}_{\alpha,\beta}}{\omega^4 + \eta^2 \omega^2} + \frac{8\pi e^2}{\Omega N_{\mathbf{k}} m^2} \sum_{n \neq n'} \sum_{\mathbf{k}} \frac{\hat{\mathbf{M}}_{\alpha,\beta}}{\epsilon_{\mathbf{k},n'} - \epsilon_{\mathbf{k},n}} \left[ \frac{(\epsilon_{\mathbf{k},n'} - \epsilon_{\mathbf{k},n})^2 - \omega^2}{[(\epsilon_{\mathbf{k},n'} - \epsilon_{\mathbf{k},n})^2 - \omega^2]^2 + \Gamma^2 \omega^2} \right] f(\epsilon_{\mathbf{k},n}) \quad (29)$$

$\eta$  accounts for the broadening due to interactions of free carriers with the external field. It represents the intrinsic damping or scattering that affects the free electrons [14].

$$\hat{\mathbf{M}}_{\alpha,\beta} = \langle u_{\mathbf{k},n'} | \hat{\mathbf{p}}_{\alpha} | u_{\mathbf{k},n} \rangle \langle u_{\mathbf{k},n} | \hat{\mathbf{p}}_{\beta}^{\dagger} | u_{\mathbf{k},n'} \rangle \quad (30)$$

This equation defines the squared matrix elements  $\hat{\mathbf{M}}_{\alpha,\beta}$ . Here,  $u_{\mathbf{k},n}$  and  $u_{\mathbf{k},n'}$  are Bloch functions, are the periodic parts of the wave functions of electrons in the crystal. They are obtained from plane-wave DFT calculations.  $\hat{\mathbf{p}}_{\alpha}$  and  $\hat{\mathbf{p}}_{\beta}$  are components of the momentum operator in the directions  $\alpha$  and  $\beta$ , respectively. The notation  $\langle u_{\mathbf{k},n'} | \hat{\mathbf{p}}_{\alpha} | u_{\mathbf{k},n} \rangle$  represents the matrix element of the momentum operator between two Bloch states. The double index  $(\alpha, \beta)$  indicates that  $\hat{\mathbf{M}}$  is a tensor. This reflects that the dielectric function is a tensor, meaning it can vary with direction in the material. Local-field effects, which consider the microscopic variations in the electromagnetic field within the crystal, is not implemented in this calculation. This can affect the accuracy of the dielectric function of those with complex internal structures.

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